

Sammith Belur

Curriculum Vitae

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RESEARCH INTERESTS

Optimization (theory and applications), Compressed Sensing, Numerical Methods, Dynamical Systems

EDUCATION

Bachelor of Science in Mathematics - Honors

Purdue University
Expected May 2027

2023 – 2027

- **Selected Coursework:** Honors Real Analysis I & II, Numerical Analysis, Linear Programming and Optimization, Probability Theory, Dynamical Systems & Ordinary Differential Equations, Linear Algebra II, Complex Analysis, Honors Algebra, and Signals and Systems for Mathematicians.
- **Honors:** Dean's List and Semester Honors.

Visiting Student

Stanford University

Summer 2025

- Completed **EE 364A – Convex Optimization I** and **EE 263 – Linear Dynamical Systems**.
- Awarded a \$5,500 merit-based scholarship to attend Stanford Summer Session.

RESEARCH AND ACADEMIC EXPERIENCE

Undergraduate Research Fellow

Purdue University, Department of Mathematics

Advisor: Prof. Xiangxiong Zhang

September 2025 – Present

- Investigating exact and iterative optimization algorithms for **total variation compressed sensing**, with applications to signal and image reconstruction.
- Developing and analyzing one and two-dimensional formulations based on the **Langlois–Darbon differential-inclusion algorithm**.
- Benchmarking differential-inclusion methods against ADMM, PDHG, and parametric max-flow-based approaches, with emphasis on convergence, computational efficiency, and structure
- Preparing an expository manuscript on optimization methods for total variation compressed sensing, emphasizing the mathematical foundations and connections between differential inclusions, operator-splitting methods, and graph-based approaches
- This project and research work was supported by the **Joel Spira Undergraduate Summer Research Award** in Summer 2026

Undergraduate Course Assistant

Purdue University, Department of Mathematics

Faculty Mentor: Prof. Aaron Yip

August 2026 – Present

- Supporting instruction in **MA 421: Linear Programming and Optimization Techniques**, helping students develop intuition for optimization models, methods, and related problem-solving techniques
- Assisting with creation of course materials - notes, homework exercises, examples, and computational tools to improve students' understanding of course material

Directed Reading Program (DRP) Mentee

Purdue University

February 2025 – Present

- Studying calculus of variations and optimal control theory through an independent directed reading project, Fall 2026.
- Completed a directed reading project in knot theory and topological methods during Spring and Fall 2025.
- Presented the knot theory project in a seminar-style talk through the Purdue DRP

Undergraduate Researcher

Purdue University, Department of Mathematics

Advisor: Prof. Shuyi Weng

January 2025 – September 2025

- Studied combinatorial and algorithmic questions related to visibility graphs of polygons and tetrahedralizable polyhedra.

- Developed an interactive Visibility Graph Visualizer for computational exploration of polygon visibility graphs.

TALKS AND PRESENTATIONS

- *Exact and Iterative Algorithms for Total Variation Compressed Sensing*, Talk, Indiana Undergraduate Mathematics Research Conference, Indiana University Indianapolis, 2026.
- *Characterizing Visibility Graphs of Polygons*, Poster Presentation, Mathematical Association of America Indiana Section Meeting, 2025.
- *Knot Theory and Topological Methods*, Purdue University Directed Reading Program, 2025.

TECHNICAL SKILLS

Programming: MATLAB, Python, Julia, C/C++

Tools: Git, Linux, Jupyter, L^AT_EX, CVXPY

HONORS AND AWARDS

- **Joel Spira Undergraduate Summer Research Award**, Purdue University College of Science, 2026.
- **Stanford Summer Session Merit Scholarship**, \$5,500, Stanford University, 2025.
- **Dean's List and Semester Honors**, Purdue University.